

LectureMind AI - Complete Project Roadmap

Project: AI Meeting & Lecture Companion (LectureMind AI)

Objective: Build a web application that records or uploads lectures, converts speech to text, generates AI summaries, notes, quizzes, flashcards, reminders, and allows users to ask questions about lecture content.

Target Users: Students, teachers, professionals.

Main Modules:

1. Authentication (Student/Admin)
2. Dashboard
3. Audio/Video Upload & Recording
4. Speech-to-Text Engine
5. AI Notes & Summary Generator
6. AI Chat with Lecture
7. Quiz & Flashcard Generator
8. Assignment/Deadline Extractor
9. Search Across Lectures
10. Profile & Settings

Recommended Tech Stack:

Frontend: React, Tailwind CSS

Backend: Node.js, Express.js

Database: MongoDB

AI: Whisper API, OpenAI/Gemini API

Storage: Cloudinary/Firebase Storage

Database Collections: Users, Lectures, Transcripts, Notes, Quizzes, Flashcards, Reminders.

Development Roadmap:

Week 1-2: UI/UX design, GitHub, database schema.

Week 3-4: Login, dashboard, upload module.

Week 5-6: Speech-to-text integration.

Week 7-8: AI summaries and notes.

Week 9: AI Q&A.;

Week 10: Quiz & flashcards.

Week 11: Reminder extraction, search.

Week 12: Testing, deployment, documentation.

Future Scope: Live transcription, Zoom/Google Meet integration, multilingual support, mobile app, analytics dashboard.

Resume Highlights: AI, NLP, Full Stack Development, REST APIs, Authentication, Cloud Storage, MongoDB, React, Node.js.